

Reference Books:      1. Introduction to Software Project Management by Adolfo Villafiorita  
                                 2. Software Engineering by Ian Sommerville

## Feasibility Study in Software Engineering

A feasibility study is the initial step in the software development life cycle (SDLC) that evaluates whether a proposed software project is practical, achievable, and worth investing in.

- To assess the viability of a software project
- To identify potential risks and constraints
- To support decision-making — whether to proceed, modify, or cancel the project

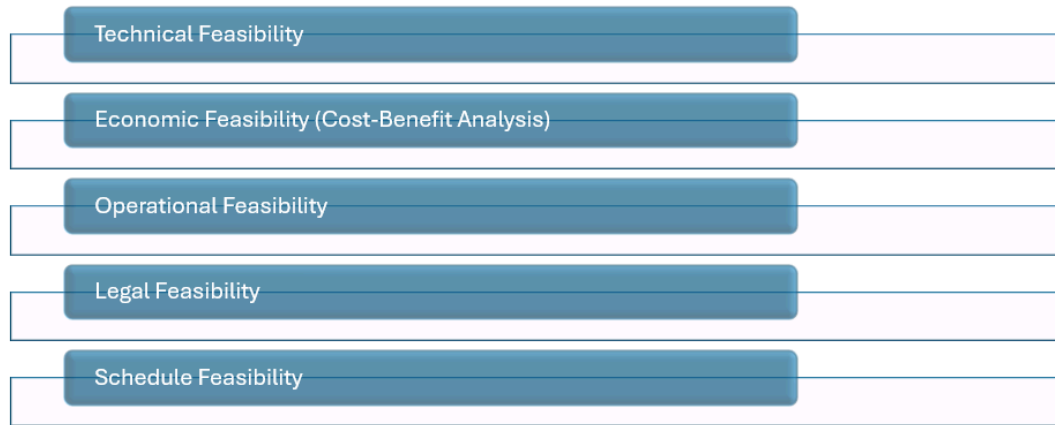


Figure 1: Steps in Conducting a Feasibility Study

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| <p><b>Outcome: Feasibility Report</b></p> <p>A <b>feasibility report</b> summarizes findings and gives a clear <b>go/no-go decision</b>.</p> | <p>It includes:</p> <ul style="list-style-type: none"> <li>• Project overview</li> <li>• Objectives and scope</li> <li>• Findings for each feasibility type</li> <li>• Cost-benefit and break-even analysis</li> <li>• Risks and mitigation strategies</li> <li>• Recommendation</li> </ul>                                                                                                               |
| <p><b>Example</b></p> <p>Suppose a company wants to automate its manual attendance system:</p>                                               | <ul style="list-style-type: none"> <li>• <b>Technical:</b> Suitable biometric devices and software are available</li> <li>• <b>Economic:</b> Project cost = \$50,000; expected savings = \$20,000/year</li> <li>• <b>Operational:</b> Staff can easily use it after short training</li> <li>• <b>Legal:</b> No privacy law issues</li> <li>• <b>Schedule:</b> Can be developed within 3 months</li> </ul> |

## Types of Feasibility

In software engineering, feasibility is generally analyzed in **five main areas**:



**Technical Feasibility:** Evaluates whether the technology, tools, and skills required for the project are available and capable of developing the proposed system.

Focus

- Availability of hardware and software resources
- Compatibility with existing systems
- Technical skills of the development team
- System performance requirements (speed, storage, reliability, scalability)
- Technology risk (e.g., new, untested tools)

Questions to Ask

- Do we have the technology and expertise to build this system?
- Can the current IT infrastructure support it?
- Are there any technical risks (e.g., hardware limits, integration issues)?

Example: A company wants to implement an AI-based defect prediction tool. If the team has experience with Python and ML frameworks, and the infrastructure can handle large datasets

### **Economic Feasibility (Cost–Benefit Analysis)**

Determines whether the project is financially worthwhile. It compares expected costs and expected benefits to ensure a good return on investment (ROI).

Focus

- Development cost: salaries, equipment, software licenses
- Operational cost: maintenance, hosting, updates
- Tangible benefits: revenue, reduced labor cost, faster operation
- Intangible benefits: user satisfaction, reputation, data accuracy
- Break-even point (BEP): when benefits = costs

Methods

- Cost–Benefit Ratio (CBR)
- Net Present Value (NPV)
- Break-even Analysis

Example: Cost to develop = \$100,000 Expected annual saving = \$40,000 Break-even in 2.5 years → Economically feasible.

### **Operational Feasibility**

Checks whether the system will work in the real-world environment and be accepted by end users.

Focus

- How well the system fits into daily business operations
- User attitude, motivation, and adaptability
- Organizational changes required
- Training and support requirements
- Ease of use and reliability

Questions to Ask

- Will users accept and use the system?
- Does it solve real operational problems?
- Will it require major workflow changes?

Example: If employees resist moving from manual to automated systems due to fear of job loss, the system may not be operationally feasible, even if it is technically possible.

### **Legal Feasibility**

Ensures the project complies with all legal and regulatory requirements.

Focus

- Software licensing and copyright issues
- Data privacy and protection laws (e.g., GDPR, HIPAA)
- Intellectual property rights
- Contractual obligations with clients or vendors
- Industry standards and certifications

Questions to Ask

- Does the system comply with laws regarding data collection and storage?
- Are there any licensing or patent issues?
- Will it violate any organizational or government regulations?

Example: A hospital software that stores patient information must comply with data protection laws — otherwise, it fails legal feasibility.

### **Schedule Feasibility**

Determines whether the project can be completed within the given time frame using the available resources.

Focus

- Realistic project timeline
- Task dependencies and milestones
- Availability of skilled staff
- Time for testing and deployment
- Impact of delays

### Questions to Ask

- Can the project be completed before the deadline?
- Are there enough resources to finish on time?
- What is the impact of delay on the organization?

Example: If a system must be deployed before a government audit in 4 months but needs 6 months of development

**Cost–Benefit Ratio (CBR):** CBR compares the total expected benefits of a project to its total expected costs. It tells you how much benefit you get per unit of cost.

$CBR = \text{Total Costs} / \text{Total Benefits}$

- If  $CBR > 1$  → project is profitable (feasible)
- If  $CBR = 1$  → project breaks even
- If  $CBR < 1$  → project is not financially viable

Example: Total Cost = \$100,000 , Total Benefit = \$150,000,  $CBR = 150,000 / 100,000 = 1.5$ . So, this project is economically feasible

**Break-Even Analysis (BEA):** Break-even analysis identifies the point at which total cost = total revenue (or benefits) — no profit, no loss.

## Break Even Point Formula

$$\text{Break Even Point} = \frac{\text{Fixed Cost}}{\text{Selling Price Per Unit} - \text{Variable Cost Per Unit}}$$

$$\text{Break Even Point} = \frac{\text{Fixed Cost}}{\text{Contribution Margin Per Unit}}$$



Example: Fixed cost = \$50,000, Selling price = \$10 per license, Variable cost = \$5 per license  
 $BEP = 50,000 / (10 - 5) = 10,000$  licenses. Need 10,000 sales to cover costs.

**Net Present Value (NPV):** NPV measures the current (present) value of all future cash inflows and outflows of a project. It considers the time value of money, meaning money today is worth more than the same amount in the future.

$$NPV = \sum_{t=1}^n \frac{B_t - C_t}{(1 + r)^t}$$

B<sub>t</sub> = Benefits (cash inflow) in year *t*, C<sub>t</sub> = Costs (cash outflow) in year *t*, *r* = Discount rate (interest or inflation rate), *n* = Number of years

If NPV > 0 → project is profitable

If NPV = 0 → project breaks even

If NPV < 0 → project loses money

Investment = \$100,000 , Expected benefits = \$40,000 per year for 3 years

Discount rate = 10%

Calculate the present value of each annual benefit

- **Year 1:**  $\frac{40,000}{(1 + 0.10)^1} = \frac{40,000}{1.1} \approx \$36,364$

- **Year 2:**  $\frac{40,000}{(1 + 0.10)^2} = \frac{40,000}{1.21} \approx \$33,058$

- **Year 3:**  $\frac{40,000}{(1 + 0.10)^3} = \frac{40,000}{1.331} \approx \$30,053$

Sum the present values and subtract the initial investment

NPV = 36,364 + 33,058 + 30,053 - 100,000 = -525. -----